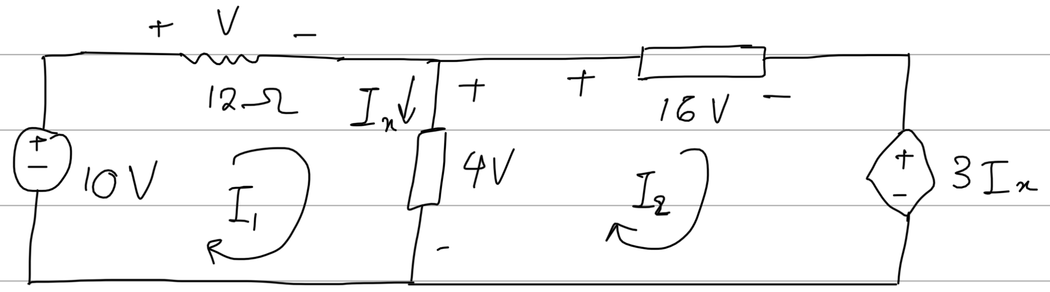
2 March

Tutorial 2

2.15 Find V and I_x



KVL Eq 2.19 $\sum_{m=1}^{M} V_m = 0$

⇒ Loop I_1

$$-10 + V + 4 = 0$$

$$V = 10 - 4$$

$$= 6\ V$$

⇒ Loop I_2

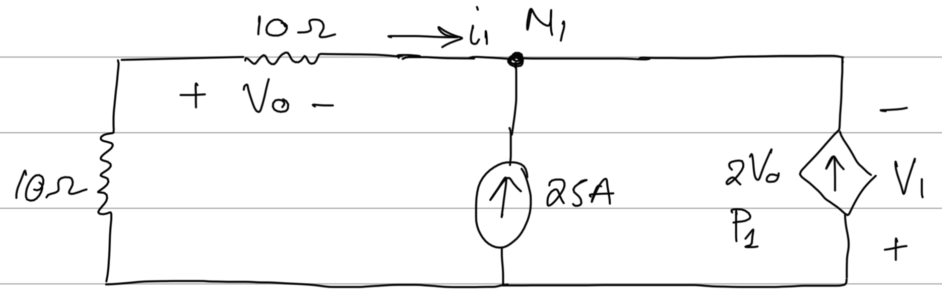
$$-4 + 16 + 3I_x = 0$$

$$3I_x = -16 + 4$$

$$I_x = \frac{-12}{3}$$

$$I_x = -4\ A$$

2.22 Find V_o and power over the $2V_o$ source



Apply KCL Eq 2.13 $\sum_{n=1}^{N} i_N = 0$

$\Rightarrow$ $V_o = i_1 R_{10\Omega}$ (1)

$i_1 + 25 + 2V_o = 0$ (2)

Substitute (2) into (1)

$\therefore$ $V_o = (-25 - 2V_o) \cdot 10$

$V_o = -250 - 20V_o$

$21V_o = -250$

$V_o = -11.904 \text{ V}$

$\Rightarrow$ $P = Vi$ Eq 1.7

$i = 2V_o$

$= 2(-11.904)$

$= -23.808 \text{ A}$

$V = i_1 R_{eq}$ where $R_{eq} = 10 + 10$

$= [-25 - 2(-11.904)](20)$ $= 20\,\Omega$

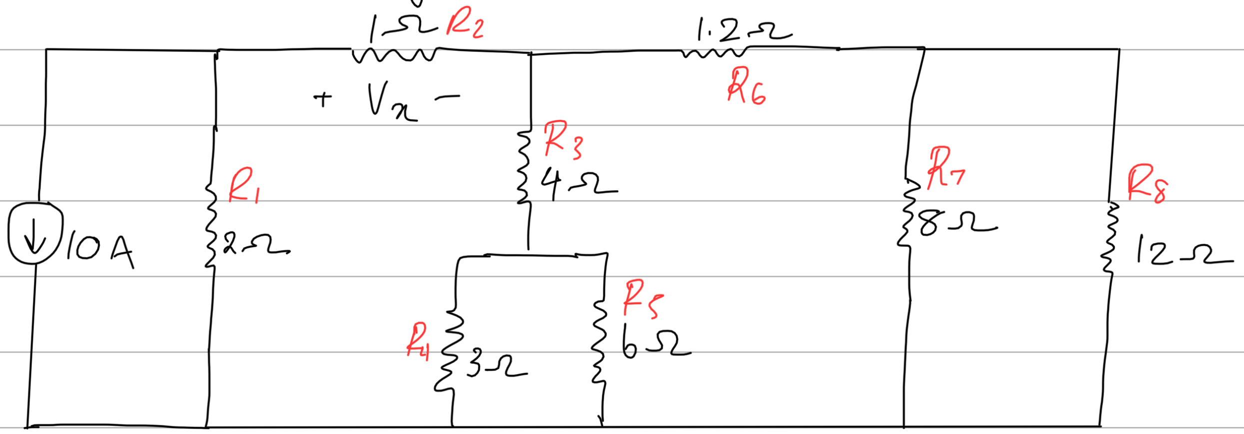
$= -23.84$ V

$\therefore P_1 = Vi$

$= (-23.84)(-23.808)$

$= 567.583$ W

2.23 In the circuit below, determine V_x and the power absorbed by the 12Ω resistor

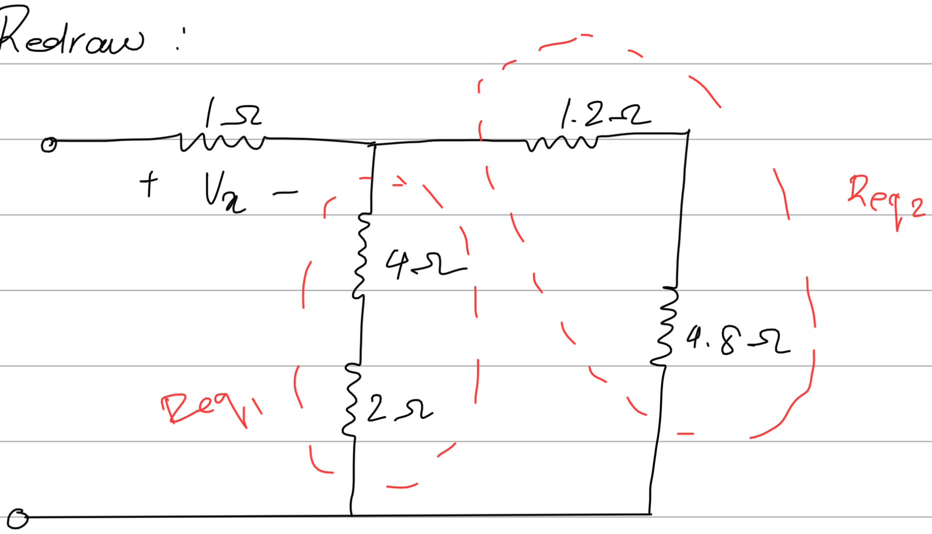


$R_7 \parallel R_8$ & $R_4 \parallel R_5$

From Eq 2.37 $\quad R_{eq} = \dfrac{R_1 R_2}{R_1 + R_2}$

$\therefore R_7 \parallel R_8 = \dfrac{8(12)}{8+12} = 4.8\,\Omega$ $\qquad R_4 \parallel R_5 = \dfrac{3(6)}{3+6} = 2\,\Omega$

Redraw :

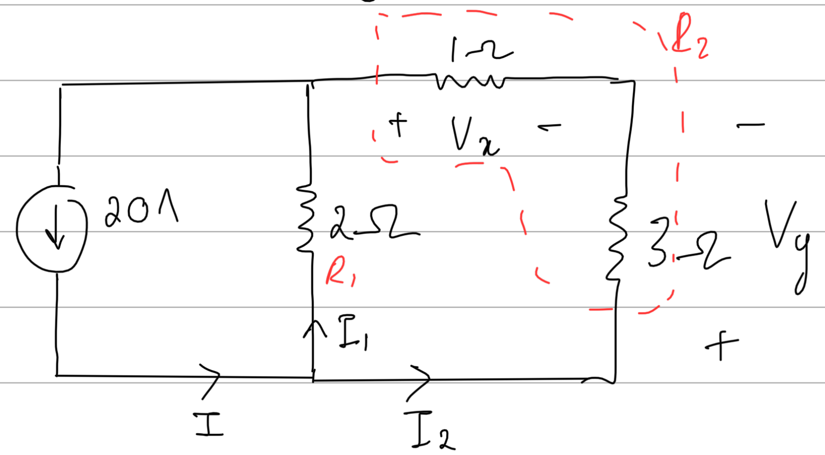


$$R_{eq_1} = 4 + 2$$
$$= 6\ \Omega$$

$$R_{eq_2} = 1.2 + 4.8$$
$$= 6\ \Omega$$

$$R_{eq_1} \parallel R_{eq_2} = \frac{6(6)}{6+6}$$
$$= 3\ \Omega$$

$\therefore$

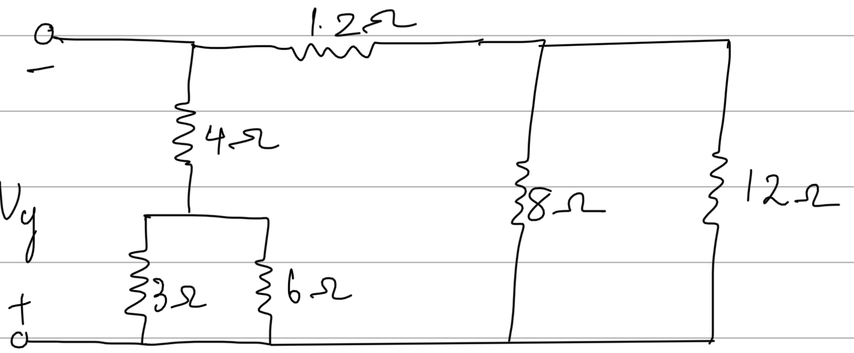


Apply current division Eq 2.43

$$I_2 = I \cdot \frac{R_1}{R_1 + R_2}$$
$$= 20 \cdot \frac{2}{2 + (1+3)}$$
$$= 6.667\ \text{A}$$

$$V_x = -I_2 R$$
$$= -6.667(1)$$
$$= -6.667\ V$$

$$\Rightarrow P = Vi = \frac{V^2}{R}$$



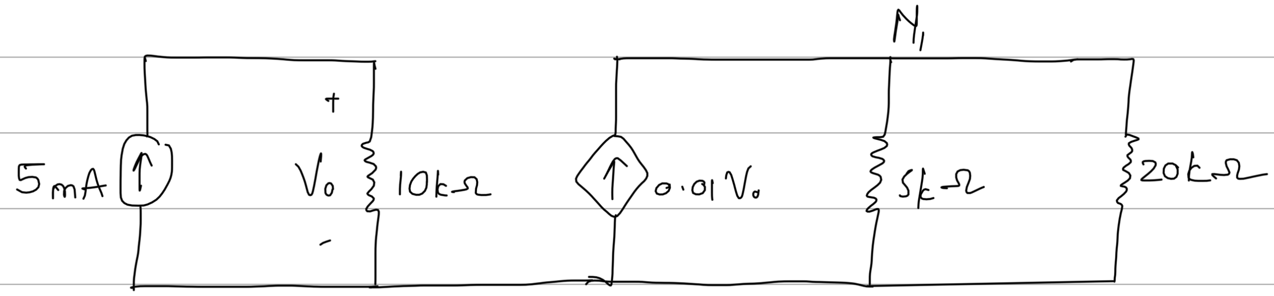
$$V_y = I_2 \cdot R$$
$$= 6.667(3)$$
$$= 20\ V$$

Voltage division

$$V_{12} = \frac{8 // 12}{8 // 12 + 1.2} \cdot 20$$
$$= 16\ V$$

$$P_{12} = \frac{16^2}{12}$$
$$= 21.33\ W$$

2.25 Find $V_{20k\Omega}$; $i_{20k\Omega}$ and $P_{20k\Omega}$



Apply current division

$$i_{20k\Omega} = i \frac{R_{5k\Omega}}{R_{5k\Omega} + R_{20k\Omega}}$$

$$i_{20k\Omega} = 0.01V_o \left(\frac{5000}{5000 + 20\,000} \right)$$

$$i_{20k\Omega} = 0.002\, V_o \quad \ldots \quad (1)$$

$$V_o = iR$$
$$= (5 \times 10^{-3})(10\,000)$$
$$= 50\ V \quad \ldots \quad (2)$$

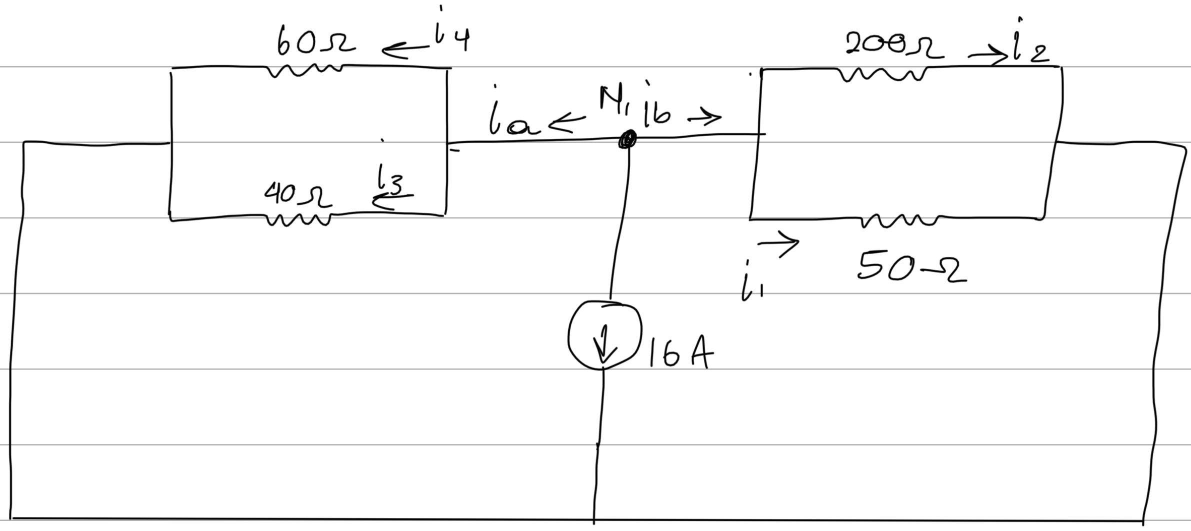
Substitute (2) into (1)

$$\therefore\ i_{20k\Omega} = 0.002\,(50)$$
$$= 0.1\ A$$
$$= 100\ mA$$

$$V_{20k\Omega} = i_{20k\Omega} R_{20k\Omega}$$
$$= 0.1\,(20\,000)$$
$$= 2000\text{ V}$$
$$= 2\text{ kV}$$

$$P_{20kV} = V_{20k\Omega}\, i_{20k\Omega}$$
$$= (2000)(0.1)$$
$$= 200\text{ W}$$

2.32 Find i_1, i_2, i_3 and i_4



$i_a = i_3 + i_4$ & $i_b = i_1 + i_2$

Redraw:

$R_a = 60 // 40$

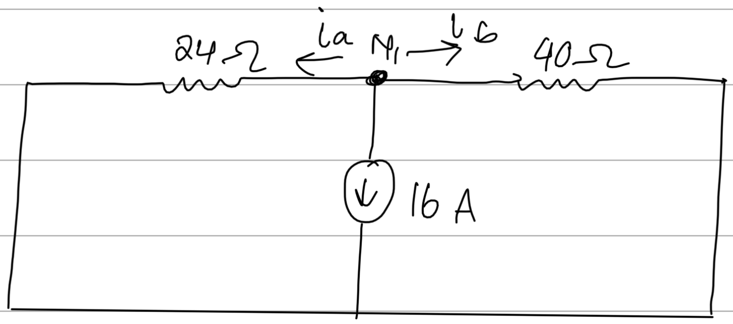
$= \dfrac{60(40)}{60+40}$

$= 24\,\Omega$

$R_b = 200 // 50$

$= \dfrac{200(50)}{200+50}$

$= 40\,\Omega$



KCL $\rightsquigarrow$ $i_a + i_b + 16 = 0$

$16 = -i_a - i_b$

$\therefore$ i_a & i_b are $-$ve

Current division

$$i_a = -i \frac{R_b}{R_a + R_b}$$

$$= -16 \cdot \frac{40}{24+40}$$

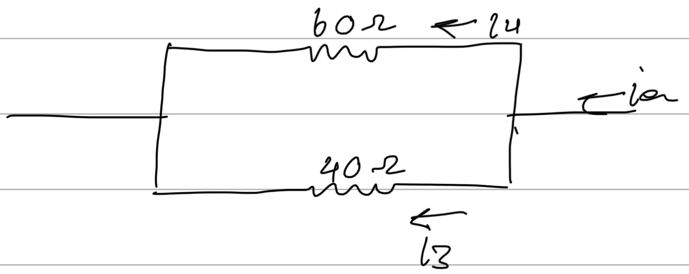
$$= -10 \text{ A}$$

$$i_b = -i \frac{R_a}{R_a + R_b}$$

$$= -16 \cdot \frac{24}{24+40}$$

$$= -6 \text{ A}$$

For i_a ; using current division



$$i_3 = i_a \frac{60}{40+60}$$

$$= -10 \cdot \frac{60}{100}$$

$$= -6 \text{ A}$$

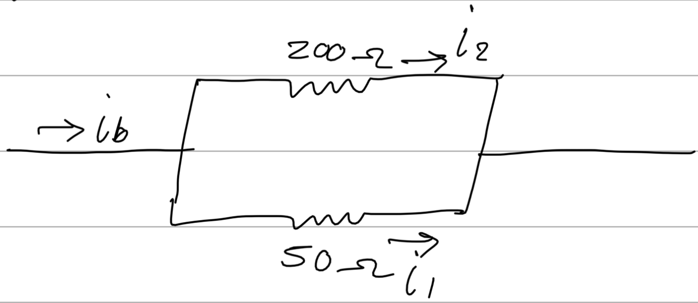
$$i_a = i_3 + i_4$$

$$i_4 = i_a - i_3$$

$$= -10 - (-6)$$

$$= -4 \text{ A}$$

For i_b



$$i_1 = i_b \frac{200}{200+50}$$

$$= -6 \cdot \frac{200}{250}$$

$$= -4.8 \text{ A}$$

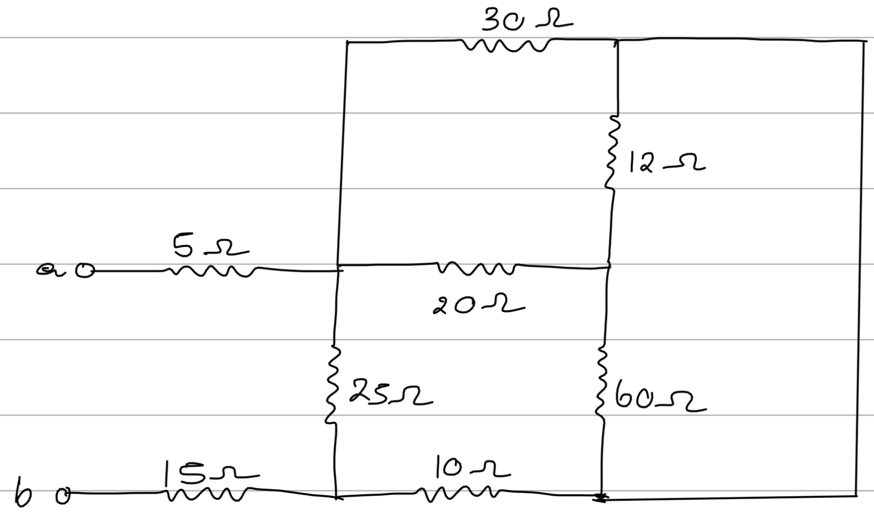
$$i_b = i_1 + i_2$$

$$i_2 = i_b - i_1$$

$$= -6 - (-4.8)$$

$$= -1.2 \text{ A}$$

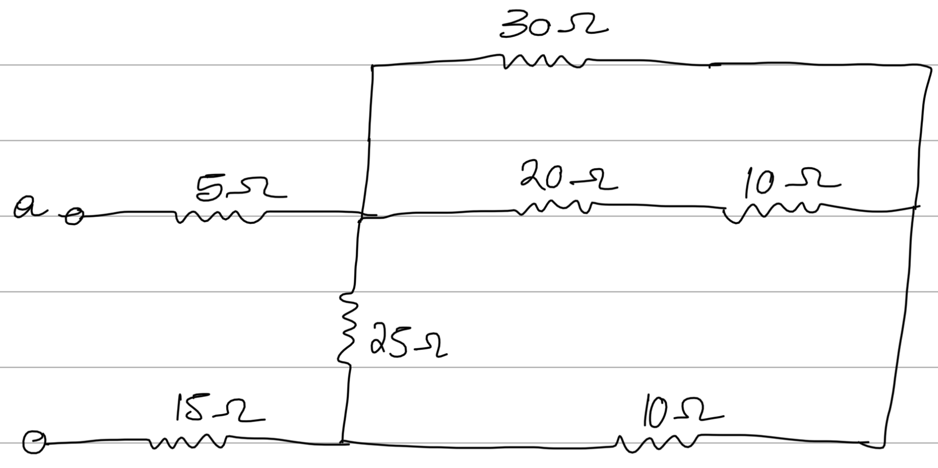
2.45b Find R_{eq} between terminals a-b



$$12 \parallel 60 = \frac{12(60)}{12+60}$$

$$= 10\,\Omega$$

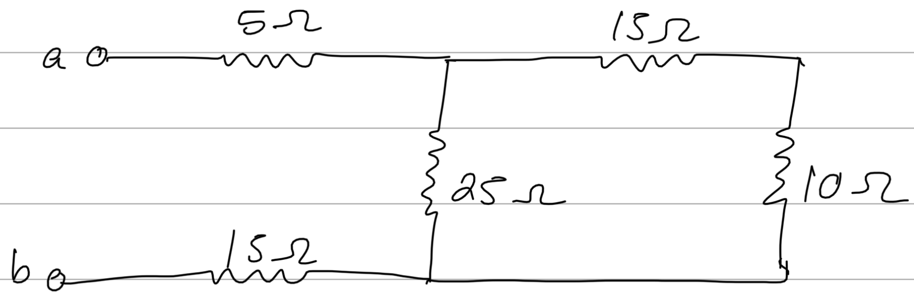
Redraw



$$30 \parallel (20+10) = \frac{30\,(20+10)}{30+(20+10)}$$

$$= 15\,\Omega$$

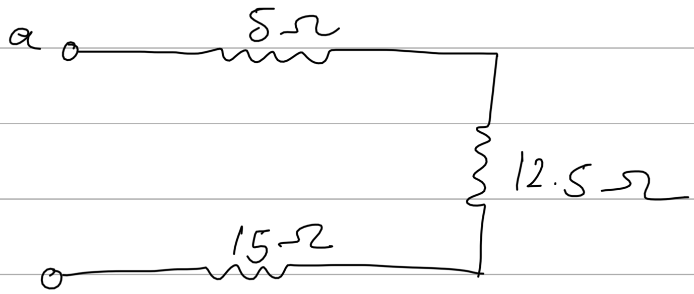
Redraw



$$25 \,//\, (15+10) = \frac{25(15+10)}{25\,(15+10)}$$

$$= 12.5\ \Omega$$

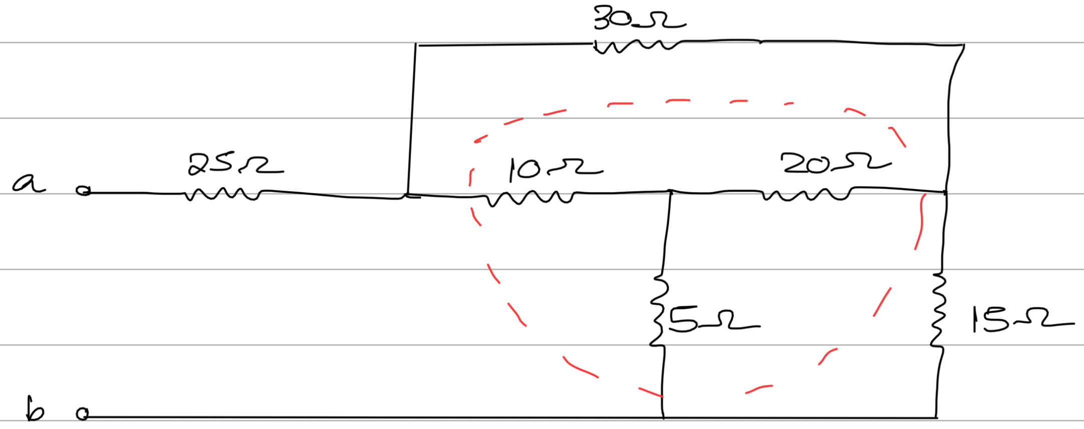
Redraw

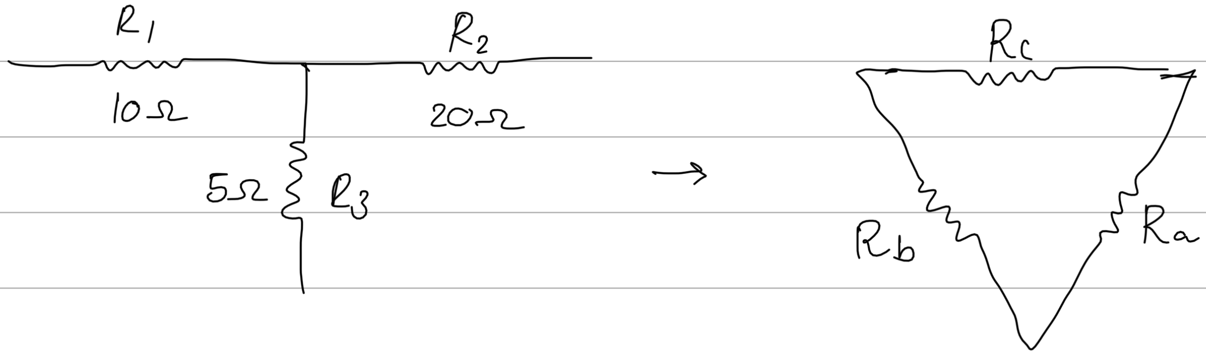


$$R_{eq} = 5 + 12.5 + 15$$

$$= 32.5\ \Omega$$

2.51 b Determine the equivalent resistance across a-b





$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$$

$= 35\ \Omega$

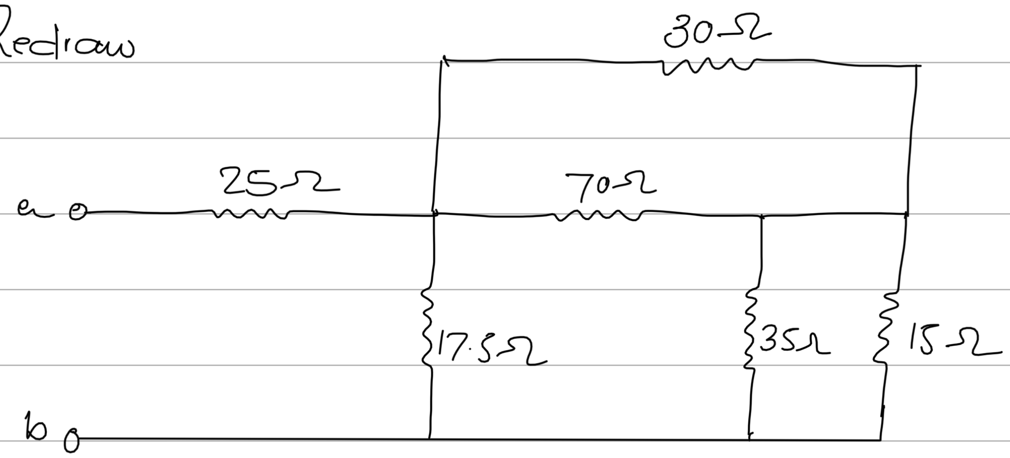
$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_2}$$

$= 17.5\ \Omega$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_3}$$

$= 70\ \Omega$

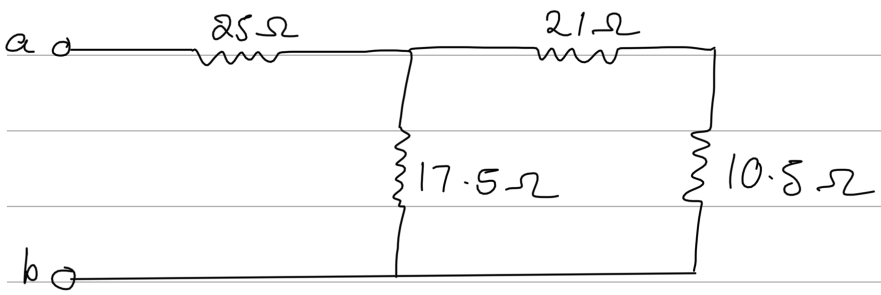
Redraw



$$35 \parallel 15 = \frac{35(15)}{35+15} = 10.5\ \Omega$$

$$30 \parallel 70 = \frac{30(70)}{30+70} = 21\ \Omega$$

Redraw



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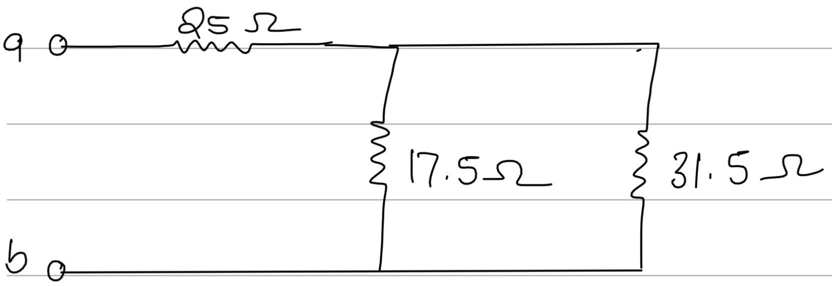
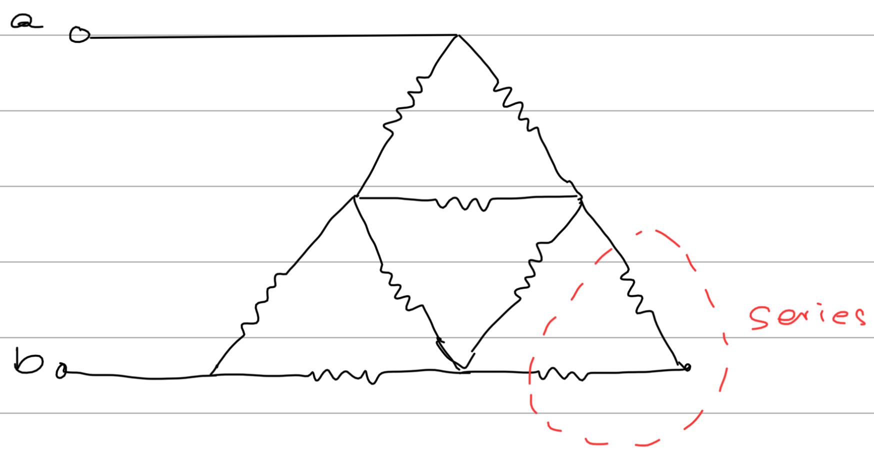


$$\therefore R_{eq} = 25 + 17.5 \parallel 31.5$$

$$= 25 + \frac{17.5(31.5)}{17.5+31.5}$$

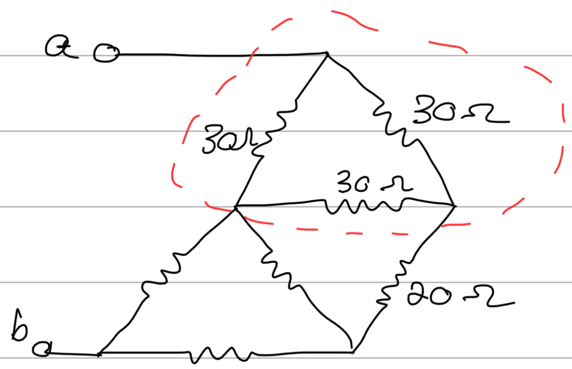
$$= 36.25\ \Omega$$

2.53b Determine the equivalent resistance across a-b if all the elements are 30 Ω



$$60 // 30 = \frac{60(30)}{60+30}$$

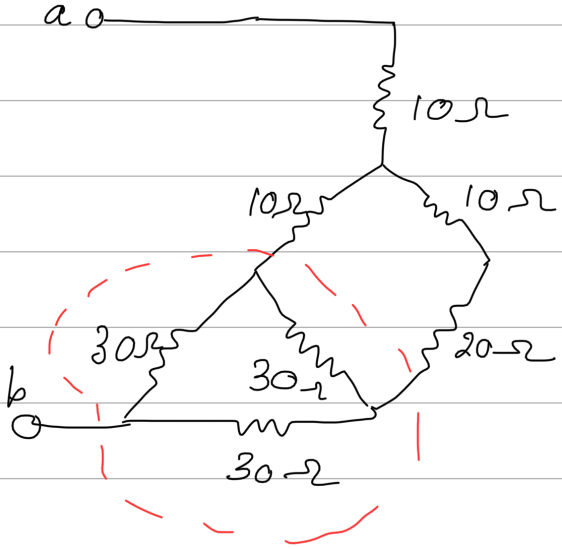
$$= 20\ \Omega$$



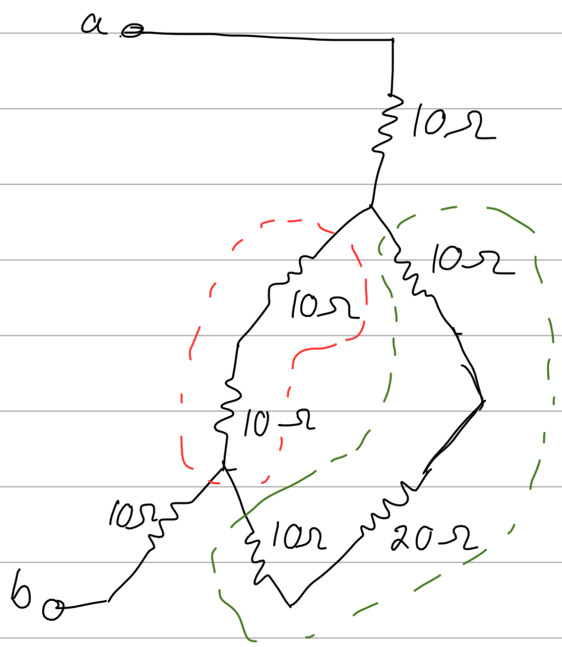
$$R_Y = \frac{R_\Delta}{3}$$

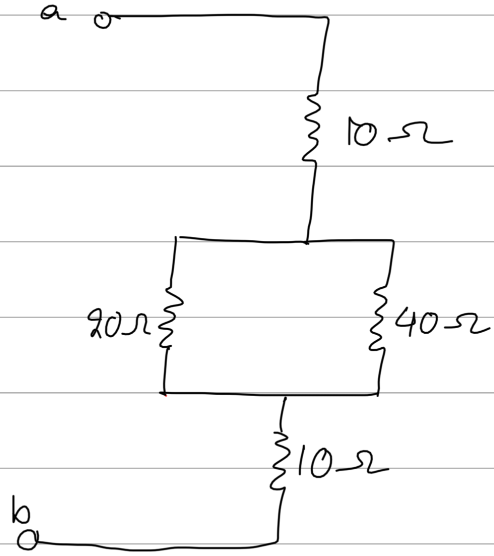
$$= \frac{30}{3}$$

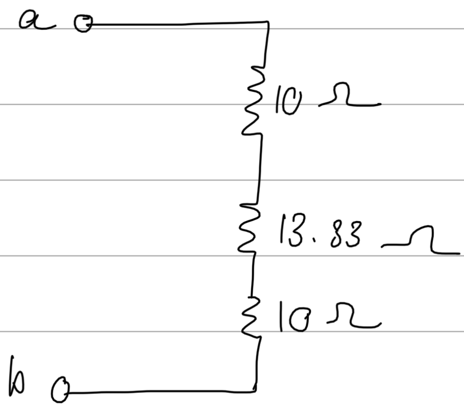
$$= 10\ \Omega$$



$$R_Y = \frac{R_\Delta}{3} = 10\,\Omega$$







$$\therefore\ R_{eq} = 10 + 13.33 + 10$$
$$= 33.33\,\Omega$$